

Samarth Khandelwal

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Education

Bachelor of Technology

Computer Science Engineering, Vellore Institute of Technology

CGPA: 9.42

2023 – 2027

Secondary & Senior Secondary

Adarsh World School, New Delhi (CBSE)

Class X (2021): **97.4%**

Class XII (2023): **92.8%**

Skills

Languages: C++, JavaScript, TypeScript

Backend & Frontend: Node.js, Express.js, React.js, REST APIs, HTML, CSS, Tailwind CSS

Databases & Tools Infrastructure: PostgreSQL, MongoDB, Redis, Git, GitHub Actions, Docker, Postman

Core Concepts: OOP, DBMS, Operating Systems, Computer Networks

Projects

Ledger Core | [Github](#)

Node.js, TypeScript, Express.js, PostgreSQL, Redis, BullMQ

- Architected a **double-entry ledger and wallet engine** with immutable, append-only ledger entries and balances derived from ledger state instead of directly mutated, enforcing atomicity across a three-phase Reserve–Execute–Finalize payment flow via PostgreSQL ACID transactions and row-level locking.
- Engineered duplicate-safe payment processing using idempotency-key hashing and conditional status transitions, with automated reconciliation to detect ledger/balance drift; validated ledger integrity across failed, rolled-back, and concurrent requests, with debits and credits remaining balanced.
- Validated correctness and performance with k6 load, spike, and soak testing, sustaining **99.44%** request success across **108,973** requests in a 20-minute, 25-VU soak test and **100%** request success at **1.81s p95 latency** under a 100-VU spike test.

CodeSentinel | [Github](#)

TypeScript, Node.js, GitHub Actions, Octokit, Gemini API

- Developed an AI-powered GitHub PR reviewer, validated on pull requests with up to **250** changed files, that analyzes code diffs and posts inline review comments.
- Designed a custom diff parsing and chunking pipeline for multi-file pull requests, enabling structured, context-aware LLM code reviews across large code changes.
- Integrated repository-aware LLM Wiki context retrieval, producing review comments with over **90% semantic similarity** to human reviews and improving cross-module impact analysis compared to baseline LLM-only reviews on live repositories.

ForeverSoles | [Github](#) [Live](#)

React, Node.js, Express.js, MongoDB, Upstash Redis, Gemini API, Stripe, Razorpay, Docker

- Engineered an atomic checkout workflow using MongoDB multi-document transactions, atomic conditional stock updates, and bounded retries (exponential backoff with jitter) to prevent overselling under concurrent purchase attempts; validated correctness using Autocannon load testing.
- Designed a cache-aside architecture using **Upstash Redis** for AI-generated Gemini review summaries with cache invalidation and latency instrumentation, reducing repeat-request latency from **1.7s to 88ms (95%)** while eliminating redundant LLM API calls.
- Implemented secure authentication and payment processing using JWT, Google OAuth 2.0, Stripe webhooks, and Razorpay HMAC verification, with CORS whitelisting and API rate limiting to ensure trusted order confirmation and secure access control.

Achievements

- Selected in Amazon ML Summer School Program 2025.
- Selected as one of 8 student representatives from the university to serve on the Academic Council, providing student feedback, academic recommendations, and inputs on university initiatives.
- Led organising committee for 20+ tech events, exhibitions and Annual Function.(2017-2023).
- Won Zonal and Inter-Zonal Debate and Declamation competitions in both English and Hindi.